

ActInf Livestream #048.2 ~ "Communication as Socially Extended Active Inference"

Livestream | 1:54:01

Hello, welcome.

It is September 14th, 2022, and we're in ACTIMF livestream number 48.2.

Welcome to the Active Inference Institute.

We're a participatory online institute that is communicating, learning, and practicing applied active inference.

You can find us at some of the links on the slide.

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Well, we are in our third discussion around this paper, Communication as Socially Extended Active Inference and Ecological Approach to Communicative Behavior from 2021 by Rémi Tisson and Pierre Poirier.

And last week in 48.1 with the very same fellows in the conversation, we had some explorations into different aspects of the paper.

Perhaps we can each say hello again.

And something, if we want to, that we remembered from last week or that we'd like to explore today and then we can just be right in it.

So, I'm Daniel.

I'm in California.

And I'm interested to look towards the latter half, the back nine of the paper and see where we go, which is kind of what we do in .2s usually.

And I just have a feeling that the exit from the slide on this one may put our communication and situatedness in a different place than where we were before.

And I'll pass it to Blue.

Hi, I'm Blue.

I heard the initial part and then lost it, Blue.

Yep.

Can you hear me now?

Yeah.

How about that?

I am Blue and I'm a researcher in New Mexico.

And I am excited to dig back into this paper.

I think we left a lot of things to wonder about last week and, yeah, excited to get to

maybe more into the meat of the paper itself as opposed to just discussing technology.

And I'll pass to Dean.

Thanks, Blue.

I'm Dean.

I'm here in Calgary.

I'm interested in seeing whether or not the big roadmap that I think that this paper provides also can be explained as a big score as in a kind of a musical analogy.

Because I'm really interested in the interplay or the interaction between orchestration and arrangements.

Yesterday, there was a math stream that where our guest was really good at sort of describing the arrangement part of what's going on in active inference potentially.

And so I'm really kind of interested in looking at that because I think from a transmission versus ecological standpoint, we're kind of looking at the difference between mono and stereo.

So I'm wondering if we should maybe look at that through that lens as well.

I'll pass it to Remy.

Hi, I'm Remy Tudon.

I'm a researcher at UCAM in Montreal, and I'm the author of the paper.

And I'm looking forward to discussing the paper more in detail and maybe go toward the later parts of the paper and maybe talk about a bit about language and linguistic communication, which is a big piece that is looming behind the paper, but it's not really that much discussed in it.

But I think that's what any account of communication aims to go towards.

So I hope we'll have the chance to discuss it a little bit.

Let's go to language.

It's what we're using among other modalities, but we are using language.

And so that sounds like a great place to start.

So what was in the paper regarding language and what was not in the paper regarding language?

Yes.

So the idea was to open a window at the end of the paper towards language.

So we talk a bit about conventional forms of communication.

We know that most people believe that language is conventional.

So we tried to tie the ideas that we put forward in the paper with work by Hampstead in a paper of 2016 about conventional affordances.

So we tried to say that the linguistic communication, the conventional nature of the language comes from the conventional affordances that are generated in the communicative interaction.

So that's what that's a way to point toward language in our account.

And we talked a bit about the acts of different types of acts of language.

How can we see these different types of speech, speech acts, as shaping the field of affordances?

What is being done ecologically with these different speech behaviors?

As opposed to the enabling aspect of the transmission view,

I think they're fairly clear within the transmission view

what these statements are about.

And as you pointed out, it's almost like hard to escape

viewing what these speech behaviors are by the content.

Like, if not their content, quite literally, then what would they be about?

Even talking about what they're about.

But informative would be sharing a proposition or reducing someone's uncertainty

about the state of something in the world.

I'm feeling this way or that weighs this much.

And so on or other statements, because indeed they're defined by the content

and their structure.

So ecologically, Remy, what might these types of speech behaviors be doing?

Oh yeah, these types of acts of language are well accounted for in the transmission view.

But there are some questions that remain about the imperative form of communication

because it's not clear that imperative communication can be understood as a form of transmission of content.

Because we have the feeling that when we command something or we request something,

we are not transmitting information or we are not aiming at changing the mental states of the person

that we are talking to, we are aiming at making them do something in the world directly.

So there's kind of a mismatch there, it seems.

So obviously, the people in the transmission view will have an answer.

They will say that like, when I have some, for instance, the desire or the intention

that something is the case, then an imperative act of communication is my transmitting this intention

or this desire to the person who then has this desire or this intention

and then can bring about the state of affairs that I intend to achieve.

But it's not, I'm not sure, obviously you can delve in the details,

but I'm not sure that's exactly what we want to do when we state,

when we produce an imperative act of language, obviously,

we don't just want that the person has the intention of doing something,

we want the thing done in the world.

So I believe that the imperative kind of communication

is not really well accounted for in the transmission view.

And what's interesting is that, in a way,

the imperative form of communication is like the basic form of communication

in our framework, because what communication is,

is acting through the behavior of somebody else

by affecting its field of affordance.

So we can go back to like the forms of animal communication that we described earlier in the paper, and these can all be considered to be imperative forms of communication or proto-imperative forms of communication.

So that's like the starting point.

And we achieve these acts on the behavior of somebody else by altering their field of affordances to make them do something.

But how do we account for informative acts of communication?

And an idea that we put forward in the paper is that the kind of acts of communication that we consider to be informative will be acts that aim to construct or alter a shared field of affordances in a way that is more durable than just an immediate imperative act of language.

So what we are doing, we are changing the layout of the context that we take to be relevant to our joint action.

And these changing of affordances do not have to be immediately acted upon.

I can say, oh, there's something in the context that's relevant.

It does not necessarily call for action immediately, but it's something that is like in the background of our context of action.

And so it changes the shared field of affordance, but it's not directly soliciting an action from the interlocutor as would be the case for the imperative active language.

So in a way, it's like as if in the beginning, an informative active language is like a long-term imperative or a dispositional imperative.

We put something in the context.

We say, oh, that's relevant.

We can act on it at some point, but it's not necessarily immediately to be acted upon.

So that's how we can put the idea that, okay, we can change our context of action, what we take to be relevant in the context of action.

And we can come back to it later when it's relevant and act on it, but it's not something that is to be acted upon immediately.

And so that's like the difference between imperative and informative begins to be discerned there where an imperative is like,

I'm saying that this affordance should be acted upon immediately.

And informative is like, I'm putting this in the context.

It changes the layout of affordances.

For instance, there would be,
I don't know how to come up with an example,
but if I inform you of something,
I'm putting, I'm changing the layout of the affordance
that we take for granted in the context of our interaction.
And it changes in the long-term, the interaction,
and it can like drive the interaction in various ways,
but it's not something to be acted directly immediately.
So that's the basic distinction.
And that's like the beginning of a distinction
between informative and imperative.
And I can say a few words about interrogative too.
So as we said,
communication is acting,
typically acting through the behavior of somebody else
by altering its field of affordance.
And interrogative,
it's interesting because it's exactly the distinction
between imperative and interrogative
is really similar to the distinction
between pragmatic value and epistemic value
in active inference.
So for instance,
when we produce an action
that aims to maximize pragmatic value,
we are trying to, for instance,
to make the state of the world
correspond to a prediction of our derivative model.
And when we make something
which is guided by epistemic value,
what we try to do
is we're trying to reduce uncertainty
about our niche, for instance.
But so in a way,
so for instance,
when I'm eating something,
I'm maximizing my pragmatic value
because I'm, for instance,

reducing a prediction error
that I'm hungry and I want to.
So that's like a basic example.
But when I'm exploring my environment,
when I'm making a visual stackade
to explore my niche,
so that would be an epistemic action,
so to speak.

So interrogative acts of language
are exactly analog to epistemic actions,
but they are epistemic actions
through the behavior of somebody else.

So instead of, for instance,
by myself exploring the niche,
exploring the field of affordance,
I can produce this act
that will make the other person
disclose to me
an element of the field of affordance
that can then reduce my uncertainty
about what is the context of our action.

So that's really the distinction
between imperative and interrogative
is really to be understood
along the line of the distinction
between pragmatic value
and epistemic value.

So that's a beginning
to an introduction
to these distinctions
in the types of active language.

Thanks, Rami.

You totally read my mind there
about like what I want to do
and I wanted to elaborate on,
even though my question
was like very poorly framed.

Yeah, I think that that's very useful

for especially in like the goal
directed aspect of imperative
versus interrogative or informative,
like what how they interact with our maybe
we don't have a joint goal
versus we do have a joint goal
or the goal is to develop
some shared state of a shared mental state.

So that was really helpful.

Thank you.

Another note is that the informative behavior
is like the base class
of the transmission view
because it seems to be like
it's updating one on the details.

Like what if I'm saying
there is a gallon of milk in the fridge,
truly what else could be conveyed
other than the content?

So it seems like the most natural
and then there are specific kinds of content
like sharing the content
that you have a question
or sharing the content
that you would like somebody to act.

And so in that way,
as you described,
many forms of speech behavior
are well accounted for
in the transmission view.

And that accounting
is actually kind of a callback
to that math stream
for with David Spivak
about accounting systems.

So these are like accounting models.

And then you highlighted
that the imperative

in the inactive
and ecological view on communication,
that the imperative
is perhaps the basic form
of communication.
And that can be seen
developmentally
through a child
and the kinds of communications
that they might engage in,
as well as from an evolutionary perspective
with the kinds of audio
or chemical
or physical stigmergy
and marks
that different kinds of creatures
use to communicate
within and across species.
So those are action-oriented
communications
as we can then say
all action-oriented communication
or all communication
we can say
is action-oriented
in an analogous way.
So then imperative
becomes the base class.
Shaping the field
of affordances,
if that's what
the ecological view
on communication is,
is going to be most natural
with kinds of expressions
that we can say
are imperative.
And then

informative updates
are like one step removed
from direct action requests.

So saying
turn the handle
to the left
is an imperative.

If you said
I'm going to inform you
about two things.

The handle can be turned
to the left
and
if this happens
you may want to turn the handle.

Those are
breaking down
a capacity for action
into
sequential
or modular
or
recomposable
statements.

But those statements
are like
hiding
the consequences
of action.

And so in that way
we can have
an action-oriented
view on communication
and then see
different kinds
of other
behavior
as like

modified

or

truncated

or censored

or hidden

action

modification.

And that may

or may not

entail

needing to even

explain

how meaning

is represented

transmitted

or received.

Yes, Remy?

Oh yeah

and

that's exactly it.

So the

that's really

the idea

explained it

very well.

And another example

that we

often use

with Pierre

is the

joint action

of cooking

a meal

with various

people.

So like

an example

we used

in the last
discussion
the example
of moving
a couch.

So another
example
maybe to
change a bit
our intuitions
about communication.

So when people
cooking a meal
together
we can suppose
that some
people
some of the
suppose that
we're in the
kitchen of
one of the
person
and the other
doesn't know
exactly where
all the things
are.

And so
the person
could say
there's flour
in the
cupboard
or in the
pantry
and saying
that that's

an informative
active language
but that
might be
a hidden
imperative
in so far
as at some
point in the
interaction
in the
cooking of
the meal
the flour
might be
needed
and at this
point
this alteration
of the
field of
affordances
this appearance
of the
flour as
an affordance
in the
context
might drive
the action
of the
person
when the
flour
is needed
at a
certain
point

the person
will have
this
affordance
in its
field
and will
go to
seek it
if it's
necessary
at that
point
so
the
informative
is
something
like
a
dispositional
or long-term
imperative
something of a
hidden imperative
that can drive
behavior
in the long-term
so
it's like
at the beginning
this distinction
is more
a distinction
of
like
is this
affordance

to be acted
upon
immediately
by you
or is it
posed
in the
context
and can
alter
the dynamic
without being
directly
acted upon
immediately
so
that's how
the basic
distinction
is drawn
and
yeah
I think
that's
how we
try to
account
for
informative
without
describing
it as
transmitting
content
or
information
so
that's

the idea
and I
think
that you
explained
it very
well
in your
in your
in your
yeah
in your
explanation
good
yeah
it's very
interesting
it's just
another
consequence
of centering
action
so many
things
the pragmatic
turn
pragmatism
earlier than
the so-called
pragmatic turn
in the
neurosciences
and all
of the
developments
in
cognitive
science

in many
cases
it's been
like
an
information
or an
inference
view
yielding
to an
action
view
and that
has just
played out
differently
on different
frontiers
and it
feels like
in a way
language
is like
the stronghold
of
information
views
because
it's like
so amenable
to syntactic
and discourse
analysis
and compositionality
and all
these features
that we see

in formal
models
and in
computer
systems
that it
feels like
oh maybe
all the funny
stuff that
people do
and say
we'll figure
that one
out
too
blue
and then
dean
so i just
wonder about
like
mimicry
which is
something that
we see
like in
the animal
kingdom
a lot
and like
whether that
is imperative
i mean
sometimes i guess
it can be
imperative
like

watch me do
this and
then you do
it
but not
always
so sometimes
it's like
you know
an animal
will see
another animal
do something
and then
they try
it too
right
and so
what
i don't know
i just don't
know how
to like
in the terms
of teaching
and learning
and like
shared goal
directedness
like what
is maybe
the role
of mimicry
and how
does that
like
what is
the effect

on the field
of affordances
does it
just open
up
like
you see
someone
do something
and then
your field
of affordances
opens
to
that
is also
possible
for me
or does
that
not
have
maybe
like
a
semantic
analog
i don't
know
yeah
that's a
really good
question
and
often
people
who
adopt

the transmission

view

will

adopt

the view

that

the main

driver

of the

evolution

of language

in human

societies

was the

transmitting

of

knowledge

and

of skills

for instance

and

of

various

types

of

activities

so

that's

like

a

clear

example

where

it

seems

that

the

transmission

view
has
a
story
to
say
when
if we
take
for
granted
that
like
imitating
somebody
or
learning
something
to
somebody
is
in
effect
transmitting
a
knowledge
or
a
certain
kind
of
mental
state
or
cognitive
state
so
there's

clearly
something
interesting
there
for
the
transmission
view
but
I
think
that
an
alternative
possible
analysis
you
just
heard
at it
is
the idea
that
you
when you're
interacting
with somebody
and
specifically
somebody
of your
own
species
seeing
this
person
do
something

can
open
new
affordances
for
yourself
for
instance
if
I
see
somebody
that
looks
like
me
do
something
I
have
learned
that
when
somebody
looks
like
me
does
something
usually
maybe
with
a
little
bit
of
effort
I

can
do
it
too
so
somebody
doing
something
is
a
way
of
opening
up
new
affordances
for me
and
in
many
cases
mimicry
will
not
necessarily
be
communicative
because
an
organism
can
do
something
without
having
no
it
is

not
communicating
nothing
to
anybody
but
it
happens
that
somebody
else
watches
it
and
imitates
it
so
in
that
case
there's
not
necessarily
communication
but
in
cases
where
there
is
communication
involved
I
think
we
can
analyze
it

just
as
you
said
that
is
saying
that
the
action
of the
organism
opens
up
a new
affordance
for
the
person
who's
watching
it
doing
it
and
usually
in
cases
of
explicit
skill
learning
for instance
the
expert
will
do
the

thing
more
deliberately
more
slowly
more
explicitly
to
show
what are
the
relevant
movements
that
have
to
be
made
in
order
to
achieve
so
that
like
making
more
salient
the
kind
of
movement
that
have
to
be
made
to

succeed
in
performing
the
skill
so
I
think
that's
a way
that we
can
understand
this
kind
of
skill
learning
in
the
transformation
of the
affordances
that are
available to
the person
that's
receiving
the
communication
that's
how I
would
I
think
I
would
analyze

it
one
simple
example
kind of
building on
this
making
certain
actions
deliberate
then
Dean
interested
for your
thoughts
even
showing
something
deliberately
verbally
calling
attention
or just
the position
of the
body
for a
physical
skill
it
shapes
the
field
of
affordances
specifically
where

somebody

can

look

so

if

you're

like

okay

I'm

going

to

swing

the

bat

and

now

I'm

going

to

show

you

what

my

hand

you

know

this

isn't

usually

how

I

do

it

but

I'm

going

to

show

you
how
I
was
holding
it
at
that
time
you
have
shaped
their
field
of
oculomotor
affordances
and
that
is
what
enables
their
generative
transmitting
that
this
is
to
be
held
between
these
two
fingers
let
alone
counterfactuals

like
here's
what
you
do
if
your
hand
is
slippery
but
by
bodily
posture
including
vocal box
bodily
posture
we
can
have
an
integrated
account
of
mimicry
skill
transfer
again
transfers
playing
way
too
strongly
into
a
transmission
view

but
skill
resonance
or
something
similar
that
helps
recognize
the
cognitive
entities
learning
process
Dean
okay
well
I want
to take
what you're
saying
but I'm
going to
put a
little bit
of a
weird
flex
on it
because
I want
to try
to combine
Blue's
pin
and
Remy's
pin

and
Daniel's
pin
into
one
kind
of
question
that
comes
back
to
the
paper
so
like
the
last
five
minutes
of
the
conversation
has
been
focused
on
the
act
of
and
as
typing
when it
pertains
to
language
and

communication

and

blue

bringing

mimicry

in

is a

form

of

as

versus

a lot

of the

emphasis

in the

paper

which is

on the

of

of

communication

and

language

so

here's

my

question

I went

back and

read the

paper

again

and

I

caught

myself

asking

what do

I
hear
when
I'm
reading
something
or
listening
to
someone
what do
I
hear
as both
processor
and the
producer
as
we
mentioned
off
camera
an
arranger
of
something
being
signaled
which I'm
picking up
on
and decoding
so
rather than
just leave it
there
I'll park
that for a

second
because I
think
that's
the
nut
of
what
I
want
to
ask
as
a
processor
and a
producer
as
an
arranger
I
can
hear
things
in
mono
I
can
hear
things
in
stereo
I
can
even
hear
things
if I

go
to
my
Apple
music
in
spatial
audio
so
that is
a field
of affordances
question
as opposed
to a
source
of
affordances
question
and
if I
only
limit
myself
to
the
source
I'm
going
to be
asking
only
questions
right
so
if I
want to
stay

in the
as
of
if I
want
to
be
able
to
hold
up
two
things
at
once
at
a
minimum
the
stereo
or
the
spatial
that's
when I
get
into
the
ecological
even
if
it's
specific
ecological
because
I'm
not
collapsing

down
to
a
single
source
as
my
point
of
contact
which
then
leads
me
to
ask
the
fact
that
communication
can
be
formed
as
of
types
meaning
mono
stereo
spatial
mimicry
interrogation
the fact
that it
can be
broken out
into
those

things
does
that
introduce
self
production
into
the
self
processing
in
your
paper
because
I know
there's
a heavy
emphasis
on
the
of
on
the
processing
but
I
wonder
if
you
keep
the
as
when
blue
brings
it
up
or

when
Daniel
brings
it
up
if
we're
keeping
it
on
that
even
a
specific
ecological
level
are we
not
when we
hear
something
producers
as well
what are
we
producing
I
have no
idea
but I
know
I hear
something
even
when
I
read
what

you're
typing
like
not
just
hearing
the
tap
I'm
hearing
something
we
are
prod
produce
we
are
producers
like
it's
changing
I'm
hearing
that
not
reading
that
so
if
I
incorporate
that
what
am
I
doing
and
I'm

not
the
expert
on
communication
so
I
wonder
what
the
expert
could
tell
me
about
that
yeah
that's
a good
question
yeah
I'm not
sure
what to
say
about
that
we'll
first
and then
we'll
continue
so
I
wonder
if
what
you're

hearing
has
anything
to
do
with
the
internal
monologue
right
so
as
you
are
reading
something
then
you
are
also
hearing
something
right
and
so
it's
like
there's
this
internal
I
mean
I
have
a
very
I'm
a

talker
person
and
so
I
have
a
strong
internal
monologue
all the
time
running
even
getting
ready
in
the
morning
I
need
to
get
my
shoes
and
get
my
purse
and
my
keys
and
I
have
to
pack
my

lunch
and
like
all
these
things
right
like
that
happen
so
I'm
constantly
hearing
that
like
as if
I'm
saying
it
maybe
to
myself
even
though
I'm
not
speaking
but
I
wonder
what
you
know
so
I
definitely
am

a
producer
of
whatever
I'm
reading
like
I
reproduce
it
I
reproduce
it
though
right
so
I
reproduce
it
in my
own
mind
my
internal
monologue
but
I
wonder
like
I
know
not
everyone
has
this
internal
monologue
and

so
so
what
is it
then
the
same
for
the
people
that
don't
perhaps
have
this
internal
monologue
I mean
maybe
they carry
around
a
picture
of
all
the
things
that
they
need
to
get
ready
in
the
morning
so
yeah

yeah
yeah
and I
think you
bring up
a good
point
is it
a
monologue
is
it
a
stereolog
is
it
a
!
the
so
but
the fact
that we
can
type
it
like
like
not
type
it
out
but
literally
turn
it
into
types

is
the
part
where
I
don't
really
have
the
background
to
be
able
to
explain
how
that
ability
that
field
of
affordances
as
opposed
to
examining
this
as
a
single
source
of
affordance
changes
the
communication
game
because

it
certainly
does
speak
to
mimicry
and
what
we
can
and
cannot
pick
up
it
certainly
speaks
to
Daniel's
point
about
what
happens
when
layers
appear
and
disappear
because
we
narrow
down
what
we're
picking
up
on
versus

have
both
of
our
ears
working
even
though
we
think
it's
our
eyes
that
are
doing
all
the
circadian
right
so
again
I'm
not
trying
to
ask
the
impossible
question
I'm
just
curious
in
the sense
producer
and
processor

and
what
that
arrangement
is
dependent
upon
what
the
what
the
because
Daniel
brought
it up
in the
last
live stream
circumstances
matter
so
but as a
hearer
I wonder
if we
know enough
about
ourselves
to
really
be
able
to
give
this
anything
more
than

as
Remy
kindly
pointed
out
in
the
paper
give
this
more
than
sort
of
a
big
roadmap
view
if we
are
taking
context
into
consideration
language
thank
you
Remy
yeah
I think
the
discussion
about
internal
monologue
is really
interesting
there's a

lot of
things that
you said
that are
really
interesting
but I
want to
go back
to the
idea
of the
monologue
and
there are
some
people
who
believe
that
language
is not
even for
communication
it's about
structuring
thought
and so
there are
people
who believe
that
language
is
first and
foremost
something
that

is a
property
of the
structure
of our
thoughts
and not
a property
of systems
of
communication
and
so
there's
this
idea
that
the
inner
monologue
comes
first
and then
comes
the
communication
of this
inner
monologue
in
communication
so
we have
this
basic
that is
once
more

inherent
in the
transmission
view
that we
have
our
train
of
thought
we
have
the
inner
speech
that we
have
in our
heads
and then
we
translate
this
inner
speech
in
overt
speech
that
is
then
taken
by
somebody
else
and
I
think

that
this
gets
the
priority
wrong
between
outer
speech
and inner
speech
and there's
been
a lot
of work
in
philosophy
by
notably
Wilfred
Sellers
and
by others
in
psychology
like
Vygotsky
who say
the first
form of
speech
is
overt
speech
that
is
then
internalized

that
becomes
inner
but
first
in the
beginning
there's
only
overt
speech
a
child
that
learns
language
does
not
have
an
inner
speech
there's
only
outer
speech
that
is
expressed
directly
the
child
speaks
the
child
do
things
with

his
words
in
its
environment
and
does
not
directly
automatically
as
the
corresponding
inner
monologue
in
its
head
and
there's
a
process
that's
really
interesting
I
think
that
is
where
we
see
that
beginning
to
happen
where
the

child
speaks
to
itself
alone
when
there's
nobody
else
around
and
so
there's
this
transition
where
language
is
used
in
communicative
interaction
and
at
some
point
the
child
use
there's
a
question
of
why
does
it
do
and

I
think
that
our
view
of
communication
is
really
interestingly
linked
with
views
of
inner
monologue
as
self-queuing
or
self-regulation
and so
the
child
can
auto-regulate
its field
of affordance
when it's
alone
when there's
nobody
else
around
and
in
time
this
outer

monologue

but

alone

monologue

becomes

overt

because

the

child

learns

to

not

necessarily

produce

the

sound

that

correspond

to

the

producing

of

these

acts

and

there's

this

integration

there's

this

internalization

of

outer

discourse

that

becomes

inner

discourse

and
there's
a lot
of
cute
anecdotes
about
that
my
nephew
is
I
have
a nephew
the son
of
my
sister
and
he's
like
when
he's
about
to
go to
sleep
he's
in
his
bed
and
every
night
he's
rehearsing
the
events

that
happen
in
the
day
and
he's
saying
the
the
comments
that
the
rules
the
norms
that
he
heard
in
the
day
oh
I
don't
have
to
do
that
I
must
do
that
!

verify
it
but

I
believe
that's
true
and
it
seems
that
I
heard
that
so
it
seems
that
first
there's
the
outer
monologue
the
outer
dialogue
and
then
there's
the
outer
monologue
and
then
there's
the
inner
monologue
that
comes
after

that
and
I
think
that's
more
the
true
direction
of
priority
between
inner
monologue
and
outer
monologue
so
I
just
have
a
cute
anecdote
about
learning
learning
to
talk
so
I
have
two
kids
and
they're
super
different

my
son
did
as
your
nephew
did
Remy
would
be
in
his
crib
talking
to
himself
talking
to
his
stuffed
animals
and
kind
of
recreating
events
in a
similar
way
he
also
babbled
a
lot
like
a
lot
of

like
just
saying
nothing
from
the
time
he
was
tiny
and
has
an
incredible
vocabulary
I
mean
now
he's
six
but
people
are
like
wow
how old
is
this
kid
I
mean
he's
reading
at
three
and
just
is

like
very
very
verbal
and
my
daughter
is
super
different
so
she
did
not
really
babble
but
she
signed
like
crazy
and
they
both
went
to
the
same
school
they
both
teach
them
the
same
signs
like
milk

and
cracker
and
cheese
and
you
know
bathroom
and
all
the
things
right
so
like
they
both
went
to
the
same
daycare
same
niche
environment
and
she
picked
up
the
signing
much
more
than
my
son
ever
did

he
used
maybe
one
or
two
signs
she
used
like
20
like
I
had
to
go
learn
the
signs
and
before
she
could
sign
she
would
stand
in
the
kitchen
and
scream
like
just
unable
to
communicate
whatever

it
was
that
she
wanted
and
it's
so
odd
like
they're
just
like
two
totally
different
kids
but
she
when
she
was
learning
Chinese
I
had
a
parent
teacher
conference
with
the
team
and
they've
both
been
learning

Chinese
now
they're
in
private
school
they've
been
learning
for
eight
years
so
when
she
was
first
starting
to
learn
she
wouldn't
speak
in
class
and
the
teacher
in
a
different
class
and
I
was
like
oh
so

it
occurred
to
me
and
because
she
didn't
babble
and
somebody
handed
her
a
lollipop
she
was
like
two
and
I
said
say
thank
you
and
she
did
the
sign
for
it
but
she
really
did
not
speak

would
not
form
clear
words
unless
she
knew
she
could
say
them
perfectly
and
so
I
sent
her
to
tutoring
with
the
Chinese
tutor
like
just
twice
three
times
and
then
it
began
like
so
she
was
like

listening

can

I

repeat

this

can

I

engage

with

this

word

correctly

and

she

was

much

more

careful

in

that

but

both

kids

as

soon

as

they

could

talk

talked

in

their

sleep

like

crazy

anyway

there's

my

cute
story
nice
stories
the
development
of the
self
cuing
and
self
speaking
is
very
interesting
even
earlier
on
in
behavioral
analysis
and
behaviorism
like
in
B.F.
Skinner's
work
there's
extensive
exploration
of
how
covert
monologues
and
covert
speech

and
societies
of
mind
and
then
I guess
more
recently
it's
become
better
understood
how
electrical
activity
in the
body
and in
the
brain
are
still
initiated
during
covert
speech
like
it's
not
an
audible
speech
but
actually
in
various
ways

it still
is
embodied
and
then
this
question
of
reading
perhaps
is
another
area
where
we're
talking
about
in
this
language
area
what
is
similar
or
different
with
listening
to
speech
participating
in
speech
audibly
is it
different
if it's
read

or
braille
how
does
the
modality
and
the
media
influence
some
of these
things
Dean
again
I
think
the
proportions
of
two
ears
to
one
mouth
is
something
that
we
need
to
kind
of
really
hold
on to
in
this

idea
of
single
source
and
monologue
versus
what
potentially
is a
stereologue
in a
field
of
affordances
as
opposed
to
a
single
source
I
don't
know
why
we've
been
I
don't
know
what
advantages
are
given
by
having
an
ability

to
hearing
stereo
but
they
must
be
better
than
having
a
single
ear
or
a
single
reception
site
and
yet
we
don't
have
two
mouths
so
again
I
think
it's
understandable
why
from
a
transmission
point
of
view

the
deliverer
standpoint
we
always
collapse
back
to
a
source
a
single
source
but
if
we
flip
that
and
now
think
about
it
from
the
receiving
standpoint
how
does
that
change
our
sense
of
what
communication
is
because

it
doesn't
change
if I
cover
one
of
my
ears
or
if
I
have
an
earbud
in
and
it
cancels
and
yet
something
fundamentally
must
change
in terms
of
communication
and
the
joint
action
even
if
it's
monologue
in
my

own
head
versus
dialogue
or
trial
or
quadrilogue
now
again
I
haven't
done
any
research
to
try
to
figure
out
what
other
people
have
done
in
terms
of
looking
at
that
but
to
your
point
though
Renee
in

terms
of
let's
get
let's
even
get
past
transmission
we're
getting
trying
to
get
past
the
idea
of
what
is
being
delivered
and
being
trying
to
gain
a
deeper
appreciation
of
what
blue
I'll
tell
you
a
little

story
I
have
a
loquacious
son
he's
in
his
30s
now
but
he's
actually
as he's
gotten
smarter
and
wiser
speaks
less
like
occasionally
he gets
out of
developing
word
salads
that
could
serve
an
entire
community
and
I
find
that

really
fascinating
I
don't
have
to
comment
on
it
it's
just
you
notice
that
as
people
know
more
they
maybe
they
sign
more
maybe
they
find
other
ways
to
make
their
point
and
it
goes
back
to
that

timing
your
take
thing
so
yeah
great
there's
so many
references
like
the true
way
is the
one
that
can't
be
spoken
of
or
those
who
do
do
and
those
who
can't
teach
all
these
kinds
of
funny
you
know
poking

fun
at
different
archetypes
and so
on
and how
that's
related
to
wisdom
the
two
ears
and
one
mouth
which
you've
been
referencing
throughout
the
conversations
and the
quadraphonic
sound
quadraphenia
I'm sure
it was
amazing
it is
true that
the
enrichment
and the
spatial
nature

of
heard
sounds
soundscapes
is
it's like
something that
can't be
turned off
like the
receiver
all of our
receiving
is bilateral
visual
nostrils
and smell
proprioceptive
is heavily
multilateral
but it
has a
coarse
bilateral
symmetry
but
the
mouth
as our
action
states
needs
generative
cohesion
perhaps
just a
just so
story

but I
don't know
of any
cases
where
there
are
now
there
are some
bodily
sounds
that are
bilateral
like
perhaps
the
vibration
of a
cricket
so it's
not that
sound
production
can't be
bilateral
it's
just that
wherever
sound
production
is localized
on the
sender
when we
highlight
content
and its

transmission
where does
that account
begin?
It has to
begin with
the content
and the
representation
in the
sender
and then
you could
ask whether
it was
fully or
partially or
deceptively
recognized
by the
recipients
potentially
when we
censor the
recipients
action
but even
calling them
a recipient
kind of
puts them
onto a
back foot
we're just
censoring the
agents
engagement
ecologically

and their
interfacing with
fields of
affordances
then we
do see
that
stereophonic
enrichment
of
everything
and no
matter how
narrow
or
how many
Shannon
bits
or
Kolmogorov
bits
it doesn't
really matter
because it's
going to be
unpacked in
a generative
model
that brings
in
aspects
that
aren't
and don't
need to be
in the
signal
and hence

aren't or
don't need
to be in
our account
of the
signaling
I just
think I'm
trying to
bring back
the inference
into the
active
inference
here
because the
active
part is
definitely
the
transmission
piece
so you
know
thank you
for
reinforcing
that
but I
do
wonder
when we
do
bring
the
inference
back
why

are
our
sensing
affordances
are
those
bilateral
forms
there
must be
some
advantage
to that
and so
is it
as Blue
pointed out
because of
as
not just
the heavy
emphasis
on the
of
what
was the
product
of my
transmission
how am
I pulling
in
this
communication
and what
advantages
are there
to

not
getting
strung
out
or
hung
up
on
the
mono
log
the
single
source
one
thing
that
makes me
think
about
is
in
caricature
disinfo
representations
of bug
vision
insect
vision
it will
look like
they're
seeing
many
separate
cells
whereas
if one

were just
to take
a breath
and think
about
their
own
visual
experience
we
don't
see
two
fields
of
view
so
it's
like
not
really
making
sense
why
an insect
would
see
multiple
so
the
visual
field
because
there's
quote
objects
out
there

we
have
all
these
different
ways
that
we
can
talk
about
how
we
know
that
we're
experiencing
an
integrated
generative
model
like
the
blind
spot
the
differences
in
visual
acuity
!
motor
saccades
and
the
way
the
attention

is
suppressed
during
circadian
so
that
it's
not
causing
this
nausea
all
these
features
are
really
important
and
they
point
beyond
a
photon
transmission
view
a
vision
into
this
generative
stereophonic
vision
and
then
in
the
audio
setting

having
multiple
oral
orifices
and
also
even
covering
them
up
like
they're
still
transduction
through
the
skull
and
so
on
we
are
also
seeking
to
flesh
out
a
generative
model
of
sounds
in
our
environment
so
if we
cover

our
ears
like
that
proprioception
will
help us
understand
that even
if
something
sounds
kind
of
quiet
we
actually
we
can
recognize
that
it's
still
a
loud
noise
or
if
we
have
one
ear
in
a
pillow
we
could
recognize

that
something
is
happening
even
though
if it
is
being
heard
differently
on both
sides
so
having
multiple
sensors
is very
compatible
with a
unified
generative
model
experience
because it
forces one
to abstract
beyond any
sensor reading
if you
have one
midline
thermometer
then
that's
your
point
estimate

for
temperature
or
you
could
have
like
a
Gaussian
blur
like
I'm
just
less
confident
about
the
temperature
further
from
my
one
central
sensor
but
maybe
it's
a
tight
Gaussian
maybe
it's
very
broad
but
this
is
the

best
estimate
why
would
I
have
any
other
estimate
centered
anywhere
else
when
you
have
two
thermometers

there's
a
generative
modeling
question
about
whether
it's
going
to
be
a
Gaussian
with
a
peak
in
the
middle
or

whether
you're
going
to
have
a
peak
more
on
one
side
and
have
it
go
through
the
other
depending
on
those
sensor
readings
but
at
that
point
you're
ending
up
with
some
distributions
which
if
we
think
of

that
distribution
as
what
is
experienced
as
like
maybe
the
product
of
the
sensing
process
that
distributions
enrichment
is
of a
different
type
and
yet
of a
different
simplicity
than
any
number
of
sensor
readings
in
terms
of
what
their

raw
information
are
so
it
seems
very
relevant
this
two
ears
one
mouth
not
just
for
listening
twice
as
much
as
we
speak
but
for
listening
twice
as
actively
as
we
speak
or
something
like
that
what
do

you
think
about
senses
Remy
yeah
I
think
that
you
explained
it
well
I
think
it's
useful
as
you
said
to
have
multiple
sensory
channels
that
can
be
active
because
I
think
that's
an
argument
from
how
we

in
its
book
in
2013
book
on
the
predictive
mind
I
think
he
says
that
if
you
have
a
tribunal
and
you're
trying
to
establish
for
instance
the
culpability
of
a
convict
you
hope
that
you
have
multiple

independent
sources
of
information
because
if
these
sources
of
information
are
influence
each
other
you
have
less
reliable
information
about
the
case
so
it's
useful
from
a
reliability
standpoint
to have
multiple
independent
sources of
information
so that
may be an
explanation
of why

we have
that may be an
explanation
of their
relative
modularity
of
perception
so we know
that there are
great limits
to that
modularity
but there is
a certain
degree of
independence
in the
different
sensory
modalities
that we
have
so
yeah
about
the
two
years
obviously
I
think
maybe
if
we
had
if
our

auditory
sensory
system
was
only
devised
to
communicative
interaction
maybe
it would
be okay
to just
have
one
here
because
we
know
who's
talking
to
us
and so
we
can
direct
our
sensory
apparatus
toward
this
communicative
signal
but
because
we
use

our
ears
not
just
to
attune
to
a
communicative
interaction
but
also
to
know
what's
in
our
surrounding
it's
clearly
more
useful
to
have
a
great
reception
from
information
anywhere
around
us
so
maybe
if
the
audition
was

only
about
communication
it would
be okay
to have
just
one
ear
but
because
we
want
to
know
as
precisely
what's
happening
there
and
at
the
same
time
know
what's
happening
there
maybe
it's
it's
it's
it has
been
more
useful
to

have
these
two
ears
two
ears
on
both
sides
so
maybe
there's
something
about
that
yeah
one
short
note
and then
Dean
is
there
are
also
animals

with
anatomy
so
that
they
can
direct
their
ears
which
is

seemingly
very
analogous
to
the
ocular
motor
circadian
it's
like
an
audio
circade
and
it
just
the
human
form
outside
of
the
ear
wiggling
which
is
maybe
slightly
vestigial
representation
of
that
but
we
have
opted
for
a

strategy
with
fixed
ear
positioning
but
a
mobile
neck
which
also
makes
circades
so
that's
kind
of
our
low
speed
audio
circading
Dean
yeah
I think
the only
advantage
I could
see
to having
two
miles
is
that
I
could
eat
the

french
fries
and
drink
the
milkshake
at
the
same
time
and
I
don't
know
what
that
would
be
like
that
was
just
overpowering
but
what
I
think
is
interesting
is
in
the
dot
dot
one
blue
brought
up

the
idea
of
how
do
we
make
sure
there
there

must
be
some
advantage
to
the
form
because
as I
said
even
if
you
cover
one
of
those
intakes
it's
not
just
a
redundancy
issue
it's
a
recursion

issue
it
creates
that
ability
to
see
context
hear
context
and
smell
context
I
don't
believe
we
would
have
those
things
and
as
I
said
I
think
I
get
overwhelmed
with
context
of
having
both
french
fries
and

milkshake

it

would

just

blow

me

up

so

but

that

that's

just

that

that's

it

this

little

self

thing

do

not

give

people

two

mouths

it

can

be

overpowering

one

aspect

might

return

us

to

the

question

of

composition
ality
it
seems
like
visual
composition
and
temporal
composition
in
music
and
in
heard
speech
has
speaking
in
terms
of
generative
models
have
highly
factorizable
or
sparsely
connected
generative
models
famously
connected
to
like
the
where
and

the
what
streams
in
visual
neuroanatomy
and that's
been explored
in some
active
inference
papers
with
whereness
and
whatness
as being
factorizable
in the
generative
model
because
it
really
is
the
case
that
different
where's
can host
different
what's
and so
it makes
sense
to
have

those
two
totally
separate
taste
and
smell
which
are
more
midlined
though
there are
two
nostrils
I'm not
totally
sure
on
whether
there's
one or
two
smell
chambers
or
anything
composition
of
chemical
bouquets
seems
quite
different
like
two
paintings
that you

like
next to
each
other
or
two
songs
in
succession
that you
like
or
it's a
little bit
different
than
two
smells
or tastes
sequentially
or composed
and so
maybe
these
more
sometimes
called
basal
sensory
or
cognitive
features
are
less
effable
and
this
kind of

relates
to
a
cognitive
stack
where
verbal
and
symbolic
communication
is
like
the tip
of the
iceberg
supported
by
biological
processes
that are
differently
or less
effable
and at
first only
pointed
towards
like
through
metaphors
and
speech
butterflies
in my
stomach
and so
on
but then

ultimately
entirely
inexpressible
maybe with
some stop
in between
like being
able to
express
through
the body
like what
it's
like to
do a
certain
bodily
position
or have
a certain
experience
or be
in a
certain
physiological
state
and
this
kind of
casts a
light on
the
transmission
view
complementing
the
ecological
view

the
transmission
view
is like
down
propagating
symbolic
communication
and seeking
for that
to apply
all the
way on
down
whereas
a more
bottom-up
and arguably
eco-evo-devo
informed
perspective
is grounded
in the
ecological
setting
and then
allows for
the
crystallization
of symbolic
processes
including
communicated
symbolic
processes
which
aren't
they

supposed
to be
the
stronghold
of the
transmission
view
but then
if
even
that
phenomena
can be
captured
within an
ecological
frame
then
there's
an
extremely
strong
basis
for
the
integrative
cohesion
of an
ecological
view
as opposed
to
an
extremely
piecemeal
and
question
begging

transmission
view
like we
talked about
semiosis
and
biosemiosis
in
live
stream
47
so
will
we
see
verbal
communication
and
behavior
as
something
that is
enabled
and
with
analogy
but not
the
same
as
cellular
transduction
or are
we going
to seek
a cellular
transduction
explanation

based on
verbal
behavior
let's
return to
some of
our
topics
all right
few areas
to go
blue
any thoughts
just
anywhere you
want
to
go
yeah
so
I
just
wanted
to
follow
up
on
what
you
said
and
I
just
think
that
the
beauty
of

the
ecological
approach
to
understanding
communication
is
that
communication
between
cells
be it
semiotic
representation
or other
and between
animals
maybe with
gestures
maybe
interspecies
communication
like you
know
you tell
your cat
come
here
and
the
cat
comes
or
dog
or
whatever
so
I

think
that
the
ability
to
model
systems
like
all
systems
is
very
important
like
if
the
transmission
view
is not
capable
of
representing
all
of
the
situations
from
cell
to
cell
communication
to
human
to
dog
communication
and
all

along
the
line
I
think
that
maybe
it's
in the
way
that
the
FEP
and
active
inference
is
scale
free
and
like
doesn't
have
I
mean
it's
across
any
hierarchical
temporal
cognitive
scale
I
think
that
the
way
that

we
view
communication
also
needs
to
be
broadly
applicable
Yes
Remy
So
yeah
that's
the
aim
but
clearly
some
people
will
say
that
sophisticated
linguistic
communication
cannot
be
accounted
for
in terms
of
ecological
communication
because
as
Daniel said
the

symbolic
communication
characteristic
of language
seems to be
too removed
from the
kind of basic
interaction
described by
the ecological
approach
so there's
this
like
there's
a lot
of work
to be
done
still
to
really
explain
the
high
level
linguistic
interactions
that we
have
in terms
of
ecological
inactive
communication
but
yeah

there's
another
example
of
communication
as you
said
we can
account
maybe
for
communicative
interaction
between
cells
but
there's
also
there's
been
work
on
communication
between
plants
and
how
plants
perceive
their
environment
in terms
of
affordances
there's
been
work
on

that
and
so
I
think
that
it's
just
to
say
that
it's
another
way
in which
we can
develop
the
ecological
approach
it
seems
that
if
we
yeah
any
organism
that
behave
so
any
organism
will
have
a
field
of

affordance
and
it
seems
that
if
it
wants
to
act
on
other
organisms
in
its
environment
it
will
act
in
affecting
its
field
of
affordance
and
it
looks
like
the
kind
of
communication
that
is
at
play
between

trees
for
instance
which
alerts
each
other
about
for
instance
threats
in
the
forest
or
sources
useful
resources
in
the
ground
so
these
kinds
of
modifications
of
the
mechanisms
by
which
these
communications
take place
but
we
can
also

see
them
as
modifications
of
the
field
of
affordances
of
trees
so
I
think
that
when
I
saw
that
I
thought
it
was
really
cool
that
the
trees
behaved
at
a
really
slower
timescale
obviously
but
like
behave

when
they
go
and
they
send
their
roots
in
certain
directions
and
things
like
that
we
can
see
it
as
them
exploring
their
environment
and
reacting
to
a
field
of
affordance
that
they
so
obviously
that's
also
controversial

because
there's
all
this
debate
about
the
cognitive
states
of
plants
and
do
they
have
consciousness
or
that
thing
but
I
think
that
we
can
say
that
they
probably
have
something
like
a
field
of
affordance
and
it

seems
that
they
can
communicate
with
each
other
via
these
modifications
of their
field
of affordance
without
us
having to
say
that
they
are
transmitting
content
which
would
be
really
weird
from
a
distant
point
of
a
plant
thank
you
so

just
like
if
you
haven't
seen
the
plant
robot
created
by
the
MIT
media
lab
check
it
out
because
the
plant
connected
it
to
a
set
of
wheels
and
a
motor
and
literally
will
wheel
itself
back
and

forth
from
window
to
window
searching
for
sunlight
so
when
you
modify
the
field
of
affordances
of
a
plant
it
will
take
advantage
of
those
affordances
for
movement
or
any
kind
of
thing
and
as
to
the
complexity

of
linguistic
communication
okay
but
also
the
complexity
of
cellular
communication
especially
in
developmental
biology
and
all
of
the
growth
factors
and
signaling
cues
and
I
mean
it's
highly
complex
and
the
recombinatorial
aspect
is
also
incredibly
crazy

I
think
it
would
take
a lot
to
convince
me
that
linguistic
communication
is
more
sophisticated
or
more
advanced
than
biological
communication
in a
developing
organism
I'm not
sold
on that
thanks
Dean
can you
help me
understand
a little
bit
more
about
the
influence

piece
in the
joint
action
if
I'm
in
Madrid
a lot
of
people
who are
trying
to get
my
attention
I'm
not
very
I'm
not
I
don't
speak
Spanish
very
well
just
barely
get
by
and
so
a lot
of
times
things
are

spoken
to
me
and
there
seems
to be
a break
in terms
of the
amount
of
sensing
I
get
and so
that
joint
action
could you
maybe
especially
in the light
of that
earlier
conference
part where
you talked
about
young
kids
they
think
it appears
they're
talking
to
themselves

but
they're
talking
to
the
world
and
maybe
they're
not
influencing
the
world
but
they
would
like
to
could
you
maybe
talk
about
that
sort
of
sensing
side
of
that
joint
action
yes
so
I
think
that
what

kids
that
talk
alone
are
doing
they're
not
they
learn
to
talk
by
talking
to
each
other
and
affecting
other
persons
but
when
they
learn
to
talk
to
each
other
what
they're
doing
they're
influencing
themselves
that's
the part

about
self-queuing
so
what
they're
doing
they're
modifying
their own
field of
affordances
with
their
inner
speech
and
that's
how
they
learn
to
self-queue
and
self-regulate
themselves
by
bringing
some
images
in mind
or
things
like
that
so
I
think
that's

what
happens
when
a
child
learns
to
speak
alone
there's
obviously
a lot
more
to say
about
that
when
so
the
question
of
different
languages
when
you're
in
another
country
and
you
don't
you
so
you
you're
spoken
to
and

you're
not
understanding
correctly
but
you
you
you
pick
up
some
parts
of
what's
said
so
I
think
that
I
think
that
the
learning
of a
language
is
often
understood
as
learning
a code
so
learning
a set
of
associations
between

meanings
and
between
forms
that
can
be
either
written
or
spoken
but
I
think
that
in
the
approach
that
we're
advancing
here
learning
a
language
is
really
more
like
learning
a
skill
so
you're
learning
to
speak
and

learning
to
understand
and
interact
correctly
in a
given
language
as
you're
learning
to
do
anything
else
you're
learning
to
play
a
game
you're
learning
to
play
hockey
or
play
soccer
or
whatever
it's
the
same
kind
of
learning

and
you're
learning
to
interact
in
that
kind
of
game
and
probably
there's
a
similarity
between
various
languages
because
there's
a
similarity
I
could
talk
about
Wittgenstein
once again
but
there's
this
various
languages
resemble
each
other
because
they

have
this
family
resemblance
so
because
you
have
this
familiarity
with
speaking
in a
language
even
when
you're
in
another
place
where
you
don't
understand
the
language
you
have
a
basic
grasp
of
some
of
the
structures
of
the

game
that
is
being
played
and
so
from
the
context
you
can
get
understand
some
of
the
things
that
are
said

understand
some
of
the
things
that
are
expected
of
you
because
you
have
this
grasp
of

what
kind
of
game
is
played
when
we
speak
a
language
and
so
even
if
you're
in
another
country
when
you
don't
speak
the
language
from
the
context
you
find
yourself
you
may
be
able
to
piece
some

things
together
obviously
if
you're
in
a
place
with
absolutely
no
context
you're
in
a
white
room
completely
empty
and
somebody
is
talking
to
you
you
have
no
context
probably
you
won't
understand
anything
but
if
you're
in

a
context
where
you
can
understand
that
some
kinds
of
there's
some
goals
that
are
in
place
and
some
people
around
you
are
trying
to
achieve
some
goals
you
can
apply
your
understanding
of
the
game
that
you

play
in
your
own
language
to
this
new
language
and
so
you
can
get
some
things
you
can
understand
some
things
about
what
you
have
to
do
and
how
you
should
behave
and
you
I
think
the
idea

of
learning
a
skill
learning
how
to
play
a
game
is
a
better
analogy
to
understanding
how
you
come
to
learn
a
language
and
how
you
come
to
apply
this
knowledge
in
other
contexts
where
the
language
that

is
spoken
is
not
the
same
that
you
that
you're
must
ring
does
that
confirm
that
it
couldn't
be
content
first
that
it
would
have
to
be
context
bound
in
my
view
the
context
is
a
central
element

in
any
communicative
interaction
in
any
cooperative
communicative
interaction
I think
it's
the
context
what I
call
the
context
in the
paper
appears
as a
shared
field
of
affordance
so
that's
the
affordance
that
are
present
in
the
environmental
context
in
which

you
find
yourself
in
with
the
other
person
and
that
that's
this
element
that
often
pre-exists
the
communicative
interaction
is a
great
source
of
constraint
on
the
communicative
interaction
so
when
you
come
to
speak
with
somebody
you're
almost

never
in
an
absolutely
context
free
place
you're
always
in
a
restaurant
in
the
bus
at
the
grocery
store
you're
somewhere
and
there's
this
basic
set
of
affordances
that
are
shared
that
you
understand
that
you
can
do

this
you
cannot
do
that
and
this
set
of
constraints
offered
by
this
shared
field
of
affordances
really
drives
and
constrains
how
the
communicative
interaction
will
unfold
so
that
I
think
that
even
if
you're
in
another
country

often
the
context
will
be
hard
to
pick
up
but
sometimes
there
will
be
shared
context
that
you
can
get
a
grasp
on
and
that
can
lead
to
an
interaction
that
is
optimal
I
brought
it
in
the

physical
Rosetta
stone
as
a
special
case
of
low
context
asynchronous
communication
one
can
imagine
there
would
be
a
million
different
contexts
by
which
the
analysis
of the
stone
could
have
been
accelerated
such
as
a
person
reading
any

of
them
the
rules
or
descriptions
being
enacted
or
a
labeled
image
or
any
number
of
contexts
that
we
would
use
to
interpret
and
act
upon
the
text
that
we
read
in
this
setting
where
we
were
historically

removed
from
the
creation
then
there
still
was
residual
context
and
it
turned
out
that
that
was
enough
to
interpret
the
symbolic
materials
but
this
is
not
like
the
base
communication
case
that
we
then
need
to
see

all
communication
and
all
learning
and
education
of
languages
and
active
inference
ontology
as being
like a
Rosetta
stone
like
the
definitions
of
words
are
not
codes
to
crack
they're
like
inside
jokes
or
ways
of
communicating
on a
sports
team

that
have
to
do
with
the
context
of
the
setting
and
asynchronous
modifications
in the
environment
especially
symbolic
asynchronous
modifications
which are
relatively
new
potentially
existing
in their
modern
form for
several
hundreds
to several
thousands
of years
depending on
how we
think about
it
symbolic
asynchronous

modification
is
a low
context
communication
whereas
even the
interpretant
of
asynchronous
symbolic
communication
still
uses
and embodies
their
context
in the
perception
so we
know that
we have
stereophonic
communication
just because
we're
stereophonic
even if
we are
getting
just a
Morse
code
symbol
but
it
is a
non-controversial

here but
interestingly
it sounds
like elsewhere
that we
would want
to think
about
communication
broadly
as being
action
oriented
and embedded
rather
than
kind of
like
reverse
engineered
from the
kinds of
artifacts
that we
have today
with quite
literally
computer
codes
and
other
kinds
of
you know
Turing
computers
and so
on

okay
I think
that you
often
explain
the
ideas
that
I'm
trying
to
convey
really
better
than
me
Daniel
I'm
not
really
fluent
in
English
so
thank
you
for
explaining
better
than
me
the
ideas
that
I'm
trying
to
say

thank
you
I
appreciate
it
and
you
wrote
the
paper
and
so
it's
unsurprising
it
seems
novel
and
I
couldn't
say it
in
any
other
way
even
when
you
speak
English
Daniel
can
still
speak
better
than
you
communicate

more
effectively
the
ideas
that
you
are
trying
to
convey
even
when
English
is
your
first
language
well
context
is
non
fungible
so
I
don't
think
that
there
can
be
a
best
or
final
communication
it's
just
like

a
best
or
final
species
or
state
for
an
ecosystem
to
be
in
and
I
guess
in
the
last
25
minutes
or
however
now
that
our
ecosystem
has
been
modified
however
it
has
been
in
these
conversations
and

through
reading
and
opening
up
threads
how
will
it
shape
our
field
of
affordances
and
how
will
we
stream
and
play
differently
Dean
so
I'm
curious
from
all
three
of
you
as
a
strategy
if
you
want
to

get
past
monologue
what
do
you
do
how
many
times
do
you
play
playing
played
so
as
an
example
like a
real
pertinent
one
right
now
the way
that the
live
streams
have
been
set
up
have
always
been
sort
of

a
pre-brief
an
introduction
with
the
authors
or
a play
and
then
played
a
debrief
piece
to
this
and
so
the
communication
is
never
left
as
a
one
off
exercise
I
just
read
the
paper
there
is
affordances
set up

to
interact
with
information
to be
able
to
receive
it
at
multiple
different
time
points
because
I
think
in
the
dot
one
you
said
we
could
look
at
it
as
slices
of
time
but
that
probably
not
accurate
in terms

of
the
dynamic
that's
in
play
so
one
of
the
things
the
active
inference
institute
also
does
is
has
a
textbook
group
now
I've
not been
a part
of the
textbook
group
and I
don't
know
whether
they
do
it
in
sets

of
three
as
well
to
create
that
kind
of
triangulation
so
that
the
interaction
that
the
communication
doesn't
become
monolithic
or
monistic
or
here
here's
here's
the
words
that
have
come
down
now
off
the
mount
accept
them

and
start
carrying
them
out
even
though
you
don't
have
a
context
right
chiseling
them
into
the
stone
doesn't
increase
their
priority
at least
if you
don't
have
the
context
so
what
I'm
curious
about
from
the
three
of
you

is
if
you
want
to
get
past
instructional
in
terms
or
if
you
want
to
get
past
transmission
is
there
a
basic
requirement
to
have
multiple
reinterpretations
or at
least
the
opportunity
to
reinterpret
like
does
it
from
a

communication

standpoint

do

we

must

we

have

reflection

and

time

for

that

do

we

have

to

have

time

built

in

for

that

or

are

we

then

bound

to

a

transmitted

state

only

anybody

can

answer

yes

blue

first

so
not
like
in
the
way
that
we
have
reflection
like
we
have
a
pre
brief
and
then
the
actual
brief
and
then
the
reflection
time
in
the
live
stream
series
so
not
in
that
way
but
I

do
think
that
in
effective
communication
there
is
like
reflection
as
an
essential
component
of
that
like
communication
when you
have
a
conversation
it
shouldn't
be
about
just
waiting
for
your
turn
to
talk
right
I
mean
there
should

be
some
reflection
of
what
the
other
person
is
saying
in
what
you're
saying
back
to
them
like
hello
how are
you
doing
today
I
want
scrambled
eggs
like
if
that
was
a
conversation
it
wouldn't
like
be
it

work
right
like
so
there
has
to
be
some
element
of
reflecting
back
what
the
other
person
is
saying
to
you
in
communication
that's
effective
it's
not
just
this
like
tower
of
disjointed
blocks
being
built
yes
Remy

yeah
I think
I think
that
what you
said
the
blue
is
interesting
the
really
important
element
of
cooperative
communication
is
that
when
you're
talking
to
somebody
when
you're
in
a
dialogue
you
have
to
concede
a
part
of
your
autonomy

you
have
to
accept
to be
regulated
by the
other
person
and
that's
I
think
where
some
people
who
like
talk
past
each
other
and
don't
listen
to
each
other
and
interact
it
happens
really
often
some
friends
talk
talk

like
it
seems
like
when
they're
waiting
they're
just
waiting
when
they're
listening
they're
just
waiting
to
say
what
they
want
to
say
so
these

have
to
pay
attention
to
what
the
other
person
is
saying
and

to
have
a
proper
to
have
a
correct
communicative
interaction
you
have
to
concede
the
like
you
have
to
accept
that
your
field
of
affordance
is
regulated
by
the
person
who's
talking
to
you
you
have
to
let

yourself
be
influenced
be
constrained
by
the person
who's
talking
to
you
just
because
you
have
to
listen
to
it
and
you
you
have
to
like
it's
it's
really
just
accepting
considering
a part
of
your
autonomy
and
that's
another

idea
from
DiPaolo
as
an
active
approach
of
language
where
like
there's
in
communicative
interaction
in
any
cooperative
interaction
but in
communicative
interaction
in
particular
you
have
to
like
inhibit
a part
of
your
field
of
afford
dance
to
participate

in
this
interaction
and
listen
because
suppose
that
what
I'm
saying
is
really
boring
to
you
and
you
just
want
to
go
eat
your
piece
of
cake
in
the
fridge
like
you
have
to
in
order
to
participate

to
maintain
the
interaction
you
have
to
inhibit
this
this
solicitation
and
to
let
me
regulate
your
field
of
afford
dance
and
so
there's
this
part
of
like
I
concede
something
in
order
to
maintain
the
interaction
and

when
there's
no
more
interest
or
I
don't
want
to
be
regulated
and
there's
another
solicitation
outside
the
interaction
that's
more
interesting
to
me
but
then
I
will
leave
the
interaction
but
to
maintain
the
interaction
you
have

to
have
a
capacity
of
inhibition
and
executive
some
form
of
what
would
be
called
executive
control
to
be
able
to
listen
and
take
what
the
person
is
saying
if
you
don't
have
that
you're
just
going
to

think
about
something
else
and
not
let
yourself
be
constrained
by the
communicative
acts
of the
other
person
so
back
to
Dean's
point
about
multi-part
and
multiplayer
communication
and
engagement
when
we only
have
one
communication
then
it
can
feel
like

a
transmission
like
the
Rosetta
stone
transmitted
from
the
past
we
don't
get
a
dot
two
or
dot
one
on
the
Rosetta
stone
structuring
learning
groups
and other
activities
and
relationships
and
cooperative
communication
as
something
unfolding
bumping
into

each
other
twice
says
something
about
those
particles
that
bumping
together
just
once
doesn't
it's
almost
like
what do
they say
once is an
accident
twice is a
coincidence
three times
is a
party
or
something
like that
so
especially
with
three
because
one
two
three
there

may
be
a
enough
of a
recurrent
engagement
and
interweaving
to support
reflection
development
to
de-emphasize
any
single
of those
events
because
none of
them are
even
half
of the
total
series
of
three
so
none
of them
can even
be said
to be
a
plurality
or a
majority

if
the
dot
zero
one
and
two
were
to
vote
to vote
one of
them
out
or
something
like
that
and
I
think
it
does
point
towards
the
reality
of our
continued
interaction
with
materials
and
then
towards
structures
that
allow

us
to
engage
naturally
Remy
I
think
that
what
communicating
in
there
repetitively
with
the same
people
what it
achieves
it
it
obviously
shapes
the
generative
model
it
aligns
them
in
some
ways
so
we
come
to
share
expectations
and

share
fields
of
afford
dances
so
it
clearly
eases
communicative
interaction
because
we don't
have to
communicate
as much
when
there's
this
trade-off
between
if you
have a
lot
of
shared
context
you
don't
have
to
communicate
as
much
and
if
you
don't

have
a
lot
of
shared
context
you
have
to
communicate
more
to
have
a
useful
interaction
to
have
a
productive
interaction
interaction
and
so
what
I
think
we
have
done
is
we
have
aligned
our
generative
models
in

ways
that
helps
us
understand
each
other
better
than
we
did
at
the
beginning
for
instance
we
can
refer
to
things
that
we
said
in
the
past
we
can
say
oh
when
you
said
that
so
I
think

that's
relevant
now
and
so
we
can
point
to
relations
to
things
that
we
said
earlier
so
there's
this
continual
easing
of
communication
when
you
have
repetitive
communicative
interaction
with
somebody
that's
that
I
think
is
why
we

can
come
to
believe
that
people
we're
living
with
we
can
read
each
other's
mind
without
talking
because
there's
so much
shared
context
that
something
can
happen
and
it
can
make
us
think
something
that
is
shared
without
us

even
saying
anything
so
there's
this
context
come
to
be
really
complex
and
really
deep
and
so
yeah
interacting
repetitively
obviously
eases
communication
okay
so
we
can
meet
multiple
times
do
preparation
engagement
and
reflection
before
during
and after

a
communication
and
structure
communication
so that
there are
multiple
such
kind of
peaks
and
troughs
we
can
talk
about
aligning
or
synchronizing
in the
general
synchrony
sense
not
making
identical
but
rather
coordinating
our
generative
models
through
perception
and
learning
as

you
mentioned
it
helps
us
understand
each
other
better
like
a
goal
of
a
communication
could
be
epistemic
let's
get to
know
each
other
and
it
could
also
have
an
element
of
pragmatic
value
like
let's
take
epistemic
action

to
find
out
if
a
collaboration
is
relevant
here
and
then
we're
going
to
engage
in
these
pragmatic
actions
Remy
yeah
and I
think
that
a lot
of
interaction
that
we
share
in
our
academic
and
highly
linguistic
and
advanced

societies
really
often
look
like
transmission
interaction
so
we
are
saying
things
that
we're
thinking
to
each
other
but
I
think
that
these
epistemic
or
communicative
in terms
of
transmission
these
communicative
transmission
interactions
are
the way
they are
because
they

are
explicit
aims
of
the
interaction
in which
we're
engaged
that
is
we're
transmitting
content
now
when
we're
talking
to
each
other
because
in a
way
that's
the
goal
of
the
interaction
but
in
other
contexts
it's
because
we
have

communicative
practices
that
we
have
developed
that
we
can
have
these
goals
that
can
drive
an
interaction
if
we're
at
the
grocery
store
if
we're
in
other
contexts
we're
doing
something
else
but
when
we're
in
academic
contexts

often
the goal
of
the
interactions
is
talking
about
ideas
transmitting
some
some
some
contents
transmitting
some
some
some
theories
for
instance
explaining
theories
explaining
arguments
and so
these
interactions
we
look at
them
and
we're
like
oh
that's
clearly
there's

we're
transmitting
content
when we're
talking
but the
idea
is
that
this
is
really
a
small
subset
of
the
cases
of
communicative
interaction
that
we
engage
in
in
everyday
life
and
we
take
these
as
paradigmatic
cases
of
communicative
interaction

when
they
are
really
the
pinnacle
of
the
linguistic
civilization
we're
in a
really
rarefied
context
where
like
academic
discussions
are not
what we
do
in
everyday
settings
when
we
talk
to
each
other
and
we
do
this
it's
because
it's

an
explicit
aim
of
our
interaction
now
that
we
do
this
but
in
other
cases
in
most
cases
when
we
speak
to
each
other
the
aim
is
not
to
transfer
information
or
talk
about
theories
it's
about
doing

things
together
and
now
we're
doing
something
together
we're
talking
about
active
inference
but
it
looks
as
if
it
can
be
understood
in
terms
of
content
sharing
and
content
transmission
but
that's
because
it's
the goal
of
our
interaction

it's
not
because
it's
the
nature
of
communication
I don't
know if
I'm
being
clear
yep
Dean you
had your
hand
raised
I just
I think
I don't
want to
put my
philosopher's
cap on
but I
will
just
briefly
I think
we use
language
and
communication
to save
time
it's
an

exploitation
strategy
and so
I think
when we
put a
13 week
set of
parentheses
around a
learning
moment
or opportunity
we
oftentimes
get the
cart ahead
of the
horse
we don't
we don't
see the
pragmatics
because
we're busy
transmitting
and what
ends up
happening
is the
plan
forward
that
affordance
that
exploitation
or
optimization

or
savings
of time
jumps ahead
of the
prep
the
participate
and the
permeate
now
we can
say that's
a luxury
but
if we
don't
respect
the fact
that people
maybe don't
necessarily
learn as
well
when they're
handed the
plan
as opposed
to
given an
opportunity
to build
one or
form one
themselves
through
preparation
participation

and permeation

we could

go away

from this

live stream

and each

of us

could

literally

now

design

a better

plan

as opposed

to

Remy

came to

us

with a

plan

no

what

Remy

came

to us

with

was a

sketch

and then

we

prepped

participated

and permeated

that

now

the

the

sophistication

of whatever
comes out
of that
in terms
of a
go forward
move
for each
one
of us
is
orders
of magnitude
greater
because
somebody
didn't
say
oh
well I've
only got
13 weeks
I don't
have the
luxury
to be able
to be
pragmatic
about this
so there
I'll take my
hat off
again
I don't
mean to
sermonize
but I
think when

we actually
feel what
it's like
to let
the plan
naturally
evolve
and become
a part
of the
ecology
as opposed
to driving
the ecology
the single
source
it changes
everything
and over
and over
again
every time
we have
one of
these
three set
live
streams
it
reinforces
what you
could do
and again
you don't
even have
to be
on
the

zoom
as long
as you
give over
the amount
of time
to see
and sense
what it's
like
to prep
participate
and I
think the
most important
one of all
stick around
for the
permeate
because that's
where the
real nuggets
seem to
Daniel's
metaphor
of icebergs
things float
to the
surface
when you
give it
enough time
but I
understand
there's also
a 13
week window
and I'm

paying thousands
of dollars
to sit
in your
class
and come
on
exploit the
daylights
out of
your
communication
and I
just don't
know
that people
don't realize
that they're
cutting off
their noses
despite
their
learning
so
interesting
a lot
there
permeation
letting it
pervade
our
generative
model
or
percolate
letting those
messages
be passed

letting
that ripple
or the
perturbation
take
effect
at slower
and deeper
levels
and it
is really
a challenge
when the
logistical
and
transmission
aspects
of
communication
are
scaffolded
and those
are like
the metadata
fields that
have to be
filled out
the boxes
that have to
be checked
and they
can feel
constraining
yet
simply
removing
them
is not

a positive
solution
and so
the path
of least
action
for many
learning
scenarios
is to
replicate
the prior
over the
decades
a professor
who might
teach the
same class
in the
same way
and there's
pros and
cons to
that
in fact
some of
the
most
enlivened
learning
communities
have like
a vibrancy
that transcends
an almost
ironically
static
learning

experience
like we
all read
this thing
so it's
not that
that I
think the
mode of
communication
and cooperation
and education
that we're
exploring here
means that
every single
experience
is going
to be
like
some kind
of
full
attention
transformative
depth
psychology
I think
it would
feel
more
normal
and natural
and just
like participating
in
a cool
ecosystem

that works
with that
individual's
situation
what else
can we
add
or where
can we
go
where is
active
inference
in all
of this
where do
you see
active
inference
not like
you know
what are
your intentions
with my
child
but
where is
active
inference
in your
further
research
agenda
Remy
I think
it gives
the
like

all
encompassing
framework
and
the
the
the
the
the
the
the
abstractness
of the
framework
allows us
to apply
it to
as we
said
like
communication
between
cells
and
communication
between
like
countries
or
groups of
individuals
this
multi-scale
property
of active
inference
I think
is really

useful
in the
in
in allowing
us to
describe
communicative
interaction
between
these
various
kinds
of
free
energy
minimizing
agents
so
there's
this
that's
useful
I think
that
the
yeah
the
learning
mechanisms
that
are
that
are
at work
in
active
inference
are

really
useful
also
to
explain
how
communicative
interaction
is
conducted
and how
repeated
communicative
interaction
can lead
to
a
form
of
synchronizing
of the
generative
model
as
has been
explored
by
Vasil
in a
paper
and
in
2020
I think
so
that
I think
that's

that's
really
useful
there's
a lot
of
things
that
I think
we can
take
from the
active
inference
framework
and
yeah
the
idea
that
all
action
I think
the
fundamental
intuition
is that
all
action
is
free
energy
minimization
that's
trivial
but
I think
there's

a pragmatist
insight
in
active
inference
that we
can
use
to
support
the
pragmatist
view of
communication
the idea
that
everything
that an
organism
does
it
does
it
to
minimize
the
energy
and
active
inference
is
no
different
and
communicative
active
inference
as a

form
of
active
inference
also
aims
to
minimize
the
free
energy
of
an
organism
and
so
when
an
organism
tries
finds
itself
in
a
situation
where
there's
a lot
of
free
energy
or
there's
a lot
of
error
of
prediction

it
will
try
to
engage
it
in
its
environment
in

environment
in
order
to
bring
itself
back
in
its
viable
states
and
so
all
these
ideas
when
we
apply
them
to
communication
I
think
they
lead
not

necessarily
by
implication
but
I
think
there's
a path
to
these
ideas
through
the
pragmatic
view
of
communication
the idea
that
communication
is
another
form
of
it's
just
a
mechanism
of
regulation
of
organisms
in order
to
minimize
their
free
energy

I
think
there's
this
basic
biological
understanding
of
the
behavior
of
organism

I
think
it
leads
us
more
toward
the
pragmatist
view
of
communication
rather
than
toward
the
more
informational
or
transmission
view
that
is
more
at
home

in
our
really
highly
advanced
and
complex
linguistic
interaction
and
I
think
that
this
basic
biological
framework
is
a
good
place
to
start
to
understand
communication
in
the
pragmatist
view
very
nice
answer
I'll
read a
comment
from
Stephen

and
then
we
can
give
a
thought
on
it
as
well
as
provide
some
closing
thoughts
so
Stephen
wrote
explicit
aims
of
interaction
are
important
and
can
be
invisible
thus
communication
appears
to be
based
on
transmission
should
we

identify
the
context
through
the
affordances
or
via
naming
the
water
like
the
water
we
swim
in
that
is
is
the
context
and
the
explicit
aims
apparent
via
the
affordances
or
is
there
a
special
sauce
that
flavors

the
cultural
waters
in
which
we
are
swimming
yes
dean
special
sauce
real
quick
on
that
I
think
what
maybe
did
build
on
what
Remy
just
said
I
think
perhaps
active
inference
opens up
that
space
to
the
idea

of
a
special
sauce
it
doesn't
guarantee
that
there
will
be
one
and
it
certainly
doesn't
promise
but
what
it
does
do
is
it
says
what
are
the
possibilities
within
this
not
just
what
are
we
collapsing
down

to
I
mean
throughout
this
series
I've
been
talking
a
lot
about
well
we
can't
leave
the
absolutes
the
things
that
we've
experienced
out
of
this
equation
but
I
think
what
is
that
those
relativities
those
entailments
and

those
entanglements
always
to be
determined
can
be
moved
towards
and
maybe
accepted
with
more
abundance
if we
use
active
influence
but
that's
just
my
optimistic
take
on it
do you
have
any
other
closing
words
before
we
hear
some
final
thoughts

from
the
others
excellent
okay
blue
or
Remy
with
some
final
thoughts
how
does
this
how
will
it
permeate
with
you
okay
so
I'll
go
ahead
so
yeah
I
was
really
happy
to
have
the
chance
to
talk

about
the
paper
with
you
I
think
that
one
of
the
most
important
things
that
I'll
take
home
is
that
now
I
have
a
really
salient
solicitation
to
go
see
the
plant
robot
in
action
so
like
at

the
second
that
I
close
the
zoom
meeting
I'm
going
to
go
see
that
that's
really
something
that
I
taking
from
this
discussion
but
yeah
no
it
was
really
interesting
to
see
different
perspectives
different
backgrounds
what
these

different
backgrounds
can
bring
to
a
discussion
that
is
I
think
that
was
really
enlightening
for me
and I
was
really
happy
to
be there
I'll just
say thanks
for coming
Remy
we really
enjoyed
this
discussion
and for
me
I think
what will
permeate
is
like
the

you know
everything
I think
about
now
I just
think
about
how can
I model
this
in terms
of
active
inference
and I
feel
like
this
paper
will
you know
influence
maybe
the way
that I
think
about
communication
which is
a big
part
of
modeling
active
inference
period
especially

like in
a multi-agent
dynamical
system
so I
think
I will
probably
leverage
this
ecological
approach
in the
way that
I think
about it
going
forward
so I
appreciate
the
opportunity
to learn
about it
awesome
my
closing
thought
what I'm
left with
here
is
active
inference
as
about
affordances
it is

about
the
the
content
of the
model
is about
affordances
and active
inference
is an
affordance
it allows
us to
realize
epistemic
and pragmatic
value
today
and in
the future
and in
that way
we can
see it
as
gripper
and
gripped
just like
we could
talk about
the hand
as
gripper
and
gripped
whether

it's
holding
hands
or
grabbing
an
object
and
especially
in this
dot
two
some
very
nice
points
added
on
language
as
gripper
and
gripped
and
contextualizing
this
kind of
breakaway
verbal
community
of
research
and
scholarly
education
and
academia
and

recognizing
that
breakaway
community
and skill
as
something
that's
crystallizing
out of
a much
more
inclusive
and
pragmatically
oriented
niche
and I
think
that will
be
a win-win
framing
because
it will
help ground
what can be
abstract
and lofty
discussions
and
also
elevate
and
bring
attention
and
support

to
some
areas
and
types
of
communication
that
are
absolutely
natural
in the
field of
affordance
view
and the
ecological
view
but are
not always
respected
so highly
under
a symbolic
transmission
view
and
the
closing
image
for me
is
making
the
sauce
underwater
I'm
just

imagining
it's
like
I don't
know
Spongebob
square
pans
or
someone's
trying
to
cook
and
make
the
sauce
but
we're
adding
things
and
it's
more
just
like
happening
in
this
diffusive
space
and
in a
way
all
kitchens
are
in

fluids
that's
why
we can
smell
what
we're
cooking
but
it's
different
enough
that
the
sauce
stays
in
the
action
of
active
inference
is
when
the
sauce
is
spilling
out
of
the
pan
all
right
Remy
you're
always
welcome

back
so
hopefully
you'll
share
some
developments
in this
line
good luck
with the
rest
of
your
work
thanks
all
all
right
farewell
bye
to

Thank you.